SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: **Database Management Systems**
COURSE CODE: **CSA05**

COURSE OUTCOME COVERED

CO2: *Apply the relational model, relational algebra, SQL query structures, and views to solve database querying and data manipulation problems.* (BL3, BL6)

Activity Tool : Analytical Problem Solving (High)
Assessment Tool Weightage: 35%

Dr

QUESTIONS		Total Marks: 50	
Q.No	**Question**	**Bloom's Level**	**Marks**
1	Apply relational algebra operations (σ, π, ∪, -, ×, ⋈) on the given Employee and Department relations to retrieve required records.	BL3 – Apply	**5**
2	Write SQL DDL statements to create STUDENT(SID, Name, DOB, DeptID) and DEPARTMENT(DeptID, DeptName) tables with all appropriate constraints.	BL3 – Apply	**5**
3	Formulate SQL queries using GROUP BY, HAVING, and aggregate functions (COUNT, SUM, AVG, MAX, MIN) on a given Sales dataset.	BL3 – Apply	**5**
4	Write a correlated sub-query to find employees earning more than the average salary of their own department.	BL3 – Apply	**5**
5	Apply all types of JOIN operations (INNER, LEFT OUTER, RIGHT OUTER, FULL OUTER, CROSS) on Employee and Project tables.	BL3 – Apply	**5**
6	Create a view 'DeptSalaryView' that displays department name, total employees, and average salary. Write a query on this view.	BL6 – Create	**5**
7	Construct a relational algebra expression for the following: Find names of students who are enrolled in all courses offered.	BL6 – Create	**5**
8	Write SQL DML statements (INSERT, UPDATE, DELETE) with	BL3 – Apply	**5**

	integrity constraint enforcement for a given scenario.		
9	Apply tuple relational calculus to express a query: Find all employees who work in the 'IT' department and earn more than 50000.	BL3 – Apply	**5**
10	Design a relational schema for an Airline Reservation System and write 5 SQL queries demonstrating joins, sub-queries, and views.	BL6 – Create	**5**

Code of Conduct:

I A.chaatriksai I(192571003)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.
a.chaatriksai
Signature of the student:

RUBRICS FOR CALCULATION:

Numerical Problems					
Criteria	**Weightage**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with	Mostly correct steps with	Disorganized steps; multiple	No clear process

		correct approach throughout	minor mistakes	errors	
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretatio n with minor omissions	Limited or unclear interpretation	No interpretatio n

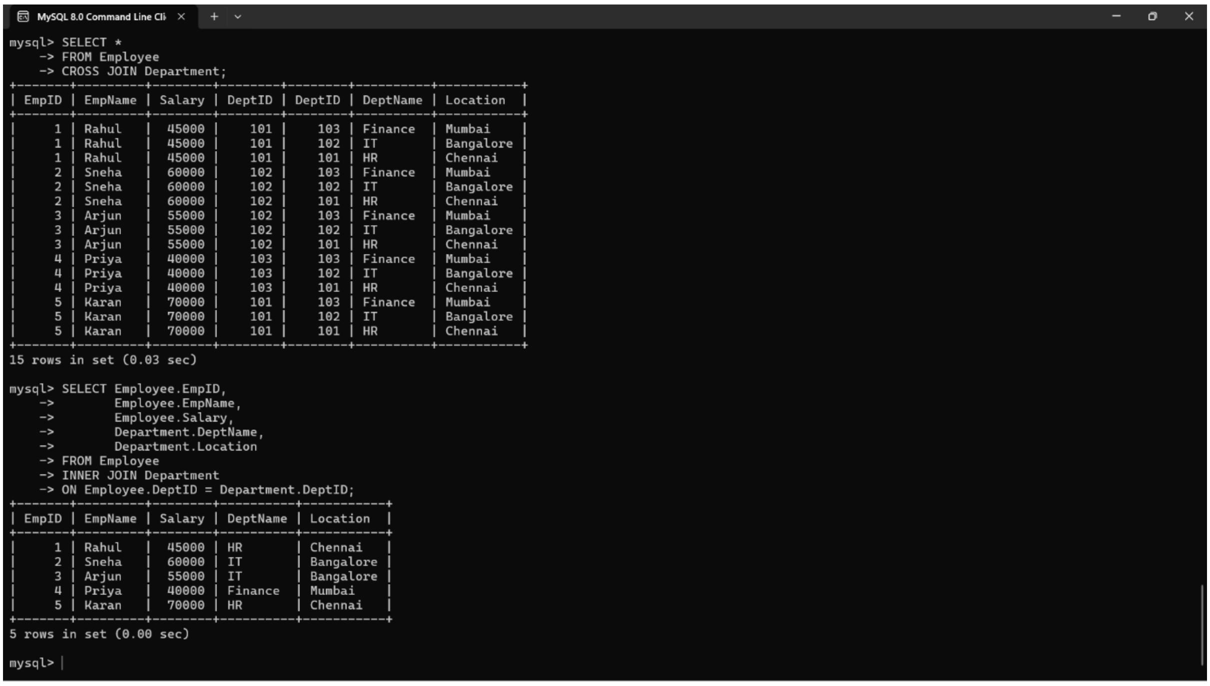
1Q)

```
create database companydb;
use companydb;

create table department(
deptid int primary key,
deptname varchar(30),
location varchar(30)
);

create table employee(
empid int primary key,
empname varchar(30),
salary int,
deptid int,
foreign key(deptid) references department(deptid)
);

insert into department values
(101,'hr','chennai'),
(102,'it','bangalore'),
(103,'finance','mumbai');

insert into employee values
(1,'rahul',45000,101),
(2,'sneha',60000,102),
(3,'arjun',55000,102),
(4,'priya',40000,103),
(5,'karan',70000,101);

select * from department;
select * from employee;

select * from employee where salary>50000;

select empname,salary from employee;

select deptid from employee
union
select deptid from department;

select deptid from department
where deptid not in(select deptid from employee);

select * from employee
```

cross join department;

select e.empid,e.empname,e.salary,d.deptname,d.location
from employee e
inner join department d
on e.deptid=d.deptid;

OUTPUT:



```
mysql> SELECT *
    -> FROM Employee
    -> CROSS JOIN Department;
+-------+---------+--------+--------+--------+----------+-----------+
| EmpID | EmpName | Salary | DeptID | DeptID | DeptName | Location  |
+-------+---------+--------+--------+--------+----------+-----------+
|     1 | Rahul   |  45000 |    101 |    103 | Finance  | Mumbai    |
|     1 | Rahul   |  45000 |    101 |    102 | IT       | Bangalore |
|     1 | Rahul   |  45000 |    101 |    101 | HR       | Chennai   |
|     2 | Sneha   |  60000 |    102 |    103 | Finance  | Mumbai    |
|     2 | Sneha   |  60000 |    102 |    102 | IT       | Bangalore |
|     2 | Sneha   |  60000 |    102 |    101 | HR       | Chennai   |
|     3 | Arjun   |  55000 |    102 |    103 | Finance  | Mumbai    |
|     3 | Arjun   |  55000 |    102 |    102 | IT       | Bangalore |
|     3 | Arjun   |  55000 |    102 |    101 | HR       | Chennai   |
|     4 | Priya   |  40000 |    103 |    103 | Finance  | Mumbai    |
|     4 | Priya   |  40000 |    103 |    102 | IT       | Bangalore |
|     4 | Priya   |  40000 |    103 |    101 | HR       | Chennai   |
|     5 | Karan   |  70000 |    101 |    103 | Finance  | Mumbai    |
|     5 | Karan   |  70000 |    101 |    102 | IT       | Bangalore |
|     5 | Karan   |  70000 |    101 |    101 | HR       | Chennai   |
+-------+---------+--------+--------+--------+----------+-----------+
15 rows in set (0.03 sec)

mysql> SELECT Employee.EmpID,
    ->        Employee.EmpName,
    ->        Employee.Salary,
    ->        Department.DeptName,
    ->        Department.Location
    -> FROM Employee
    -> INNER JOIN Department
    -> ON Employee.DeptID = Department.DeptID;
+-------+---------+--------+----------+-----------+
| EmpID | EmpName | Salary | DeptName | Location  |
+-------+---------+--------+----------+-----------+
|     1 | Rahul   |  45000 | HR       | Chennai   |
|     2 | Sneha   |  60000 | IT       | Bangalore |
|     3 | Arjun   |  55000 | IT       | Bangalore |
|     4 | Priya   |  40000 | Finance  | Mumbai    |
|     5 | Karan   |  70000 | HR       | Chennai   |
+-------+---------+--------+----------+-----------+
5 rows in set (0.00 sec)

mysql>
```

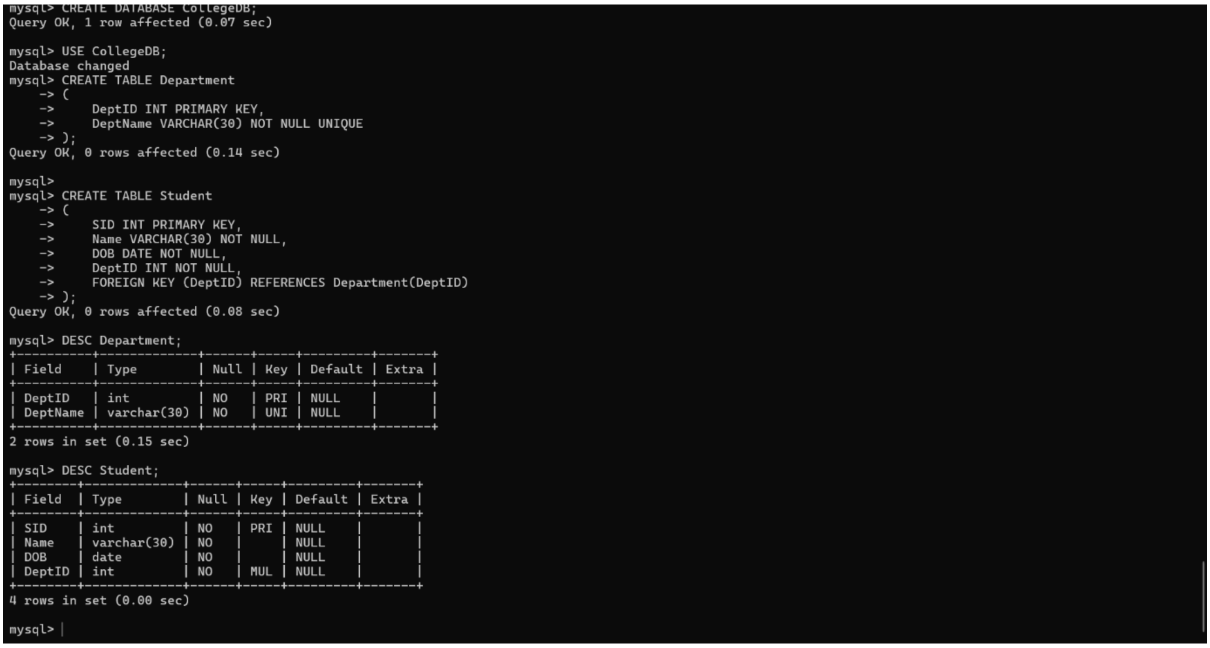
2Q)
create database collegedb;
use collegedb;

create table department(
deptid int primary key,
deptname varchar(30) not null unique
);

create table student(
sid int primary key,
name varchar(30) not null,
dob date not null,
deptid int,
foreign key(deptid) references department(deptid)
);

desc department;
desc student;

OUTPUT:



```
mysql> CREATE DATABASE CollegeDB;
Query OK, 1 row affected (0.07 sec)

mysql> USE CollegeDB;
Database changed
mysql> CREATE TABLE Department
    -> (
    ->     DeptID INT PRIMARY KEY,
    ->     DeptName VARCHAR(30) NOT NULL UNIQUE
    -> );
Query OK, 0 rows affected (0.14 sec)

mysql>
mysql> CREATE TABLE Student
    -> (
    ->     SID INT PRIMARY KEY,
    ->     Name VARCHAR(30) NOT NULL,
    ->     DOB DATE NOT NULL,
    ->     DeptID INT NOT NULL,
    ->     FOREIGN KEY (DeptID) REFERENCES Department(DeptID)
    -> );
Query OK, 0 rows affected (0.08 sec)

mysql> DESC Department;
+----------+-------------+------+-----+---------+-------+
| Field    | Type        | Null | Key | Default | Extra |
+----------+-------------+------+-----+---------+-------+
| DeptID   | int         | NO   | PRI | NULL    |       |
| DeptName | varchar(30) | NO   | UNI | NULL    |       |
+----------+-------------+------+-----+---------+-------+
2 rows in set (0.15 sec)

mysql> DESC Student;
+--------+-------------+------+-----+---------+-------+
| Field  | Type        | Null | Key | Default | Extra |
+--------+-------------+------+-----+---------+-------+
| SID    | int         | NO   | PRI | NULL    |       |
| Name   | varchar(30) | NO   |     | NULL    |       |
| DOB    | date        | NO   |     | NULL    |       |
| DeptID | int         | NO   | MUL | NULL    |       |
+--------+-------------+------+-----+---------+-------+
4 rows in set (0.00 sec)

mysql>
```

3Q)

```
create database salesdb;
use salesdb;

create table sales(
saleid int primary key,
productname varchar(30),
category varchar(20),
quantity int,
price int
);

insert into sales values
(1,'laptop','electronics',2,50000),
(2,'mouse','electronics',5,500),
(3,'keyboard','electronics',3,1000),
(4,'chair','furniture',4,3000),
(5,'table','furniture',2,7000),
(6,'pen','stationery',20,20),
(7,'book','stationery',10,150);

select * from sales;

select category,count(*) totalproducts
from sales
group by category;
```

select category,sum(quantity) totalquantity
from sales
group by category;

select category,avg(price) averageprice
from sales
group by category;

select category,max(price) highestprice
from sales
group by category;

select category,min(price) lowestprice
from sales
group by category;

select category,sum(quantity) totalquantity
from sales
group by category
having sum(quantity)>5;

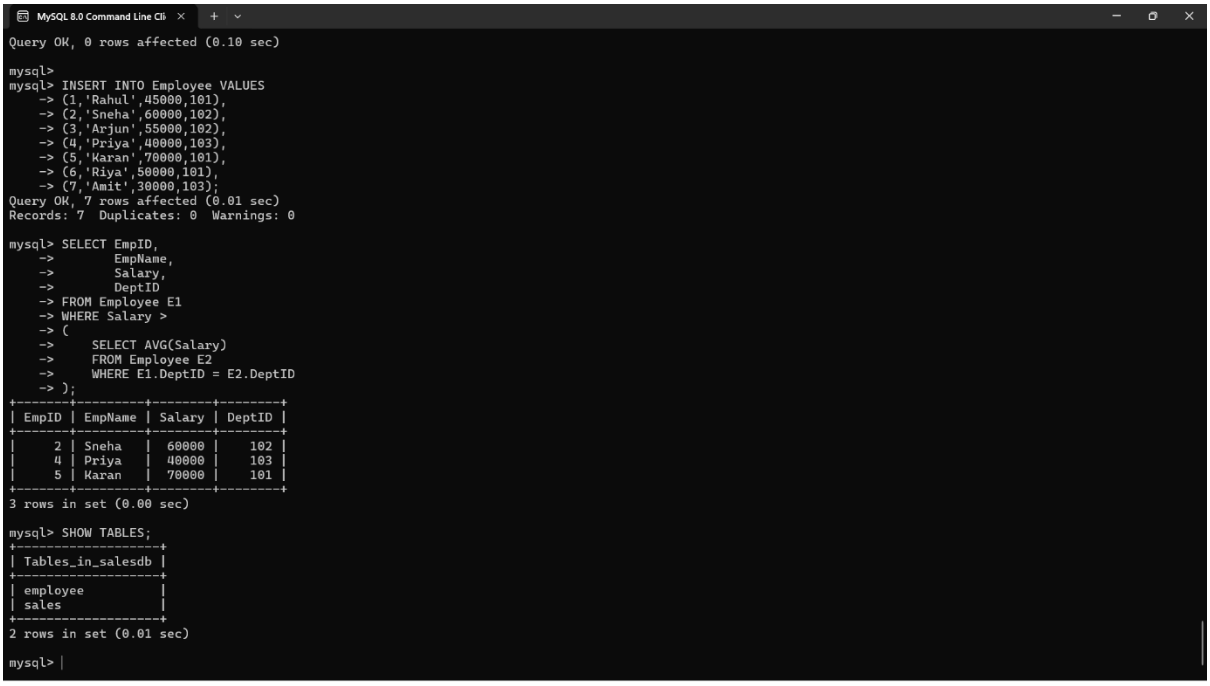
OUTPUT :

```
mysql> SELECT Category, MIN(Price) AS LowestPrice
    -> FROM Sales
    -> GROUP BY Category;
+-------------+-------------+
| Category    | LowestPrice |
+-------------+-------------+
| Electronics |         500 |
| Furniture   |        3000 |
| Stationery  |          20 |
+-------------+-------------+
3 rows in set (0.00 sec)

mysql> SELECT Category, SUM(Quantity) AS TotalQuantity
    -> FROM Sales
    -> GROUP BY Category
    -> HAVING SUM(Quantity) > 5;
+-------------+---------------+
| Category    | TotalQuantity |
+-------------+---------------+
| Electronics |            10 |
| Furniture   |             6 |
| Stationery  |            30 |
+-------------+---------------+
3 rows in set (0.03 sec)

mysql> SELECT Category,
    -> COUNT(*) AS TotalProducts,
    -> SUM(Quantity) AS TotalQuantity,
    -> AVG(Price) AS AveragePrice,
    -> MAX(Price) AS HighestPrice,
    -> MIN(Price) AS LowestPrice
    -> FROM Sales
    -> GROUP BY Category;
+-------------+---------------+---------------+--------------+--------------+-------------+
| Category    | TotalProducts | TotalQuantity | AveragePrice | HighestPrice | LowestPrice |
+-------------+---------------+---------------+--------------+--------------+-------------+
| Electronics |             3 |            10 |   17166.6667 |        50000 |         500 |
| Furniture   |             2 |             6 |    5000.0000 |         7000 |        3000 |
| Stationery  |             2 |            30 |      85.0000 |          150 |          20 |
+-------------+---------------+---------------+--------------+--------------+-------------+
3 rows in set (0.00 sec)

mysql> |
```

4Q)
use companydb;

select empid,
empname,
salary,
deptid
from employee e1
where salary>
(
select avg(salary)
from employee e2
where e1.deptid=e2.deptid
);

OUTPUT :



```
Query OK, 0 rows affected (0.10 sec)

mysql>
mysql> INSERT INTO Employee VALUES
    -> (1,'Rahul',45000,101),
    -> (2,'Sneha',60000,102),
    -> (3,'Arjun',55000,102),
    -> (4,'Priya',40000,103),
    -> (5,'Karan',70000,101),
    -> (6,'Riya',50000,101),
    -> (7,'Amit',30000,103);
Query OK, 7 rows affected (0.01 sec)
Records: 7  Duplicates: 0  Warnings: 0

mysql> SELECT EmpID,
    ->        EmpName,
    ->        Salary,
    ->        DeptID
    -> FROM Employee E1
    -> WHERE Salary >
    -> (
    ->     SELECT AVG(Salary)
    ->     FROM Employee E2
    ->     WHERE E1.DeptID = E2.DeptID
    -> );
+-------+---------+--------+--------+
| EmpID | EmpName | Salary | DeptID |
+-------+---------+--------+--------+
|     2 | Sneha   |  60000 |    102 |
|     4 | Priya   |  40000 |    103 |
|     5 | Karan   |  70000 |    101 |
+-------+---------+--------+--------+
3 rows in set (0.00 sec)

mysql> SHOW TABLES;
+-------------------+
| Tables_in_salesdb |
+-------------------+
| employee          |
| sales             |
+-------------------+
2 rows in set (0.01 sec)

mysql> |
```

5Q)
use companydb;

create table project(
projectid int primary key,
projectname varchar(30),
deptid int
);

insert into project values
(1,'website',101),

```sql
(2,'mobile app',102),
(3,'payroll',103),
(4,'ai system',104);

select * from employee;
select * from project;

select e.empid,e.empname,p.projectname
from employee e
inner join project p
on e.deptid=p.deptid;

select e.empid,e.empname,p.projectname
from employee e
left join project p
on e.deptid=p.deptid;

select e.empid,e.empname,p.projectname
from employee e
right join project p
on e.deptid=p.deptid;

select e.empid,e.empname,p.projectname
from employee e
left join project p
on e.deptid=p.deptid

union

select e.empid,e.empname,p.projectname
from employee e
right join project p
on e.deptid=p.deptid;

select *
from employee
cross join project;
```

OUTPUT :

```
mysql> SELECT E.EmpID,E.EmpName,E.DeptID,P.ProjectName
    -> FROM Employee E
    -> LEFT JOIN Project P
    -> ON E.DeptID=P.DeptID;
+-------+---------+--------+-------------+
| EmpID | EmpName | DeptID | ProjectName |
+-------+---------+--------+-------------+
|     1 | Rahul   |    101 | Website     |
|     2 | Sneha   |    102 | Mobile App  |
|     3 | Arjun   |    102 | Mobile App  |
|     4 | Priya   |    103 | Payroll     |
|     5 | Karan   |    101 | Website     |
|     6 | Riya    |    101 | Website     |
|     7 | Amit    |    103 | Payroll     |
+-------+---------+--------+-------------+
7 rows in set (0.00 sec)

mysql> SELECT *
    -> FROM Employee
    -> CROSS JOIN Project;
+-------+---------+--------+--------+-----------+-------------+--------+
| EmpID | EmpName | Salary | DeptID | ProjectID | ProjectName | DeptID |
+-------+---------+--------+--------+-----------+-------------+--------+
|     1 | Rahul   |  45000 |    101 |         4 | AI System   |    104 |
|     1 | Rahul   |  45000 |    101 |         3 | Payroll     |    103 |
|     1 | Rahul   |  45000 |    101 |         2 | Mobile App  |    102 |
|     1 | Rahul   |  45000 |    101 |         1 | Website     |    101 |
|     2 | Sneha   |  60000 |    102 |         4 | AI System   |    104 |
|     2 | Sneha   |  60000 |    102 |         3 | Payroll     |    103 |
|     2 | Sneha   |  60000 |    102 |         2 | Mobile App  |    102 |
|     2 | Sneha   |  60000 |    102 |         1 | Website     |    101 |
|     3 | Arjun   |  55000 |    102 |         4 | AI System   |    104 |
|     3 | Arjun   |  55000 |    102 |         3 | Payroll     |    103 |
|     3 | Arjun   |  55000 |    102 |         2 | Mobile App  |    102 |
|     3 | Arjun   |  55000 |    102 |         1 | Website     |    101 |
|     4 | Priya   |  40000 |    103 |         4 | AI System   |    104 |
|     4 | Priya   |  40000 |    103 |         3 | Payroll     |    103 |
|     4 | Priya   |  40000 |    103 |         2 | Mobile App  |    102 |
|     4 | Priya   |  40000 |    103 |         1 | Website     |    101 |
|     5 | Karan   |  70000 |    101 |         4 | AI System   |    104 |
|     5 | Karan   |  70000 |    101 |         3 | Payroll     |    103 |
|     5 | Karan   |  70000 |    101 |         2 | Mobile App  |    102 |
|     5 | Karan   |  70000 |    101 |         1 | Website     |    101 |
|     6 | Riya    |  50000 |    101 |         4 | AI System   |    104 |
|     6 | Riya    |  50000 |    101 |         3 | Payroll     |    103 |
|     6 | Riya    |  50000 |    101 |         2 | Mobile App  |    102 |
|     6 | Riya    |  50000 |    101 |         1 | Website     |    101 |
|     7 | Amit    |  30000 |    103 |         4 | AI System   |    104 |
|     7 | Amit    |  30000 |    103 |         3 | Payroll     |    103 |
|     7 | Amit    |  30000 |    103 |         2 | Mobile App  |    102 |
|     7 | Amit    |  30000 |    103 |         1 | Website     |    101 |
+-------+---------+--------+--------+-----------+-------------+--------+
28 rows in set (0.00 sec)
```

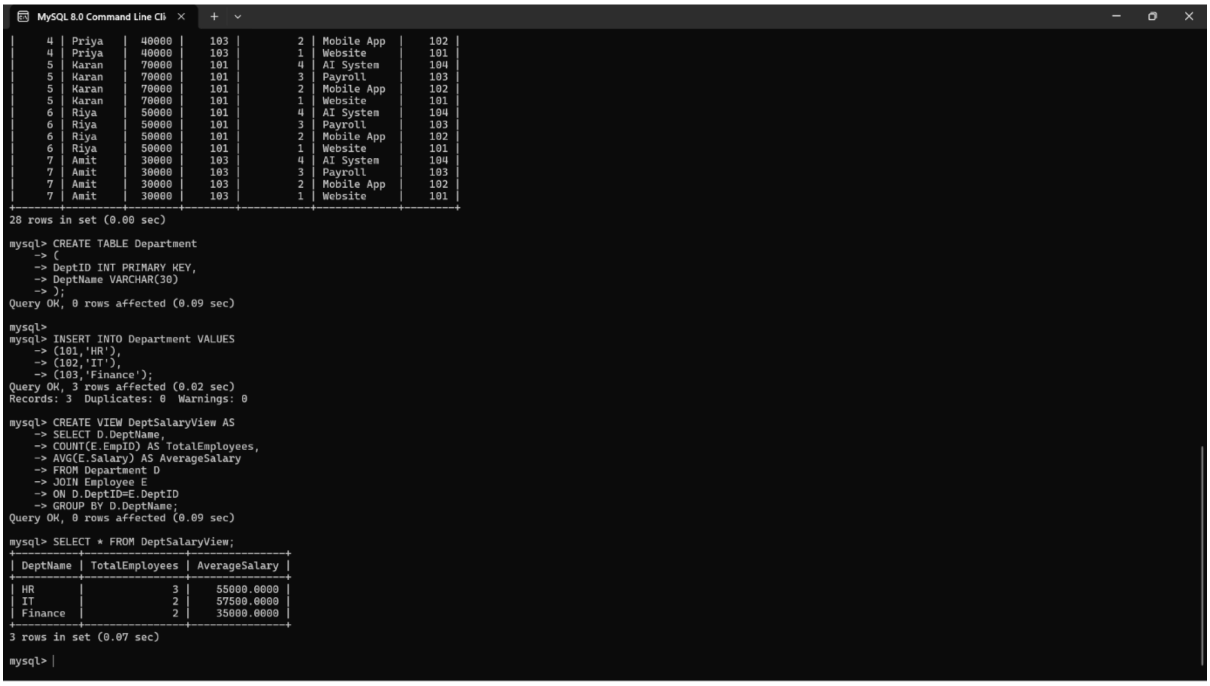
6Q)

use companydb;

create view deptsalaryview as
select d.deptname,
count(e.empid) as totalemployees,
avg(e.salary) as averagesalary
from department d
join employee e
on d.deptid=e.deptid
group by d.deptname;

select * from deptsalaryview;

OUTPUT :


```
MySQL 8.0 Command Line Cli
|     4 | Priya    |  40000 |    103 |          2 | Mobile App  |     102 |
|     4 | Priya    |  40000 |    103 |          1 | Website     |     101 |
|     5 | Karan    |  70000 |    101 |          4 | AI System   |     104 |
|     5 | Karan    |  70000 |    101 |          3 | Payroll     |     103 |
|     5 | Karan    |  70000 |    101 |          2 | Mobile App  |     102 |
|     5 | Karan    |  70000 |    101 |          1 | Website     |     101 |
|     6 | Riya     |  50000 |    101 |          4 | AI System   |     104 |
|     6 | Riya     |  50000 |    101 |          3 | Payroll     |     103 |
|     6 | Riya     |  50000 |    101 |          2 | Mobile App  |     102 |
|     6 | Riya     |  50000 |    101 |          1 | Website     |     101 |
|     7 | Amit     |  30000 |    103 |          4 | AI System   |     104 |
|     7 | Amit     |  30000 |    103 |          3 | Payroll     |     103 |
|     7 | Amit     |  30000 |    103 |          2 | Mobile App  |     102 |
|     7 | Amit     |  30000 |    103 |          1 | Website     |     101 |
+-------+----------+--------+--------+------------+-------------+---------+
28 rows in set (0.00 sec)

mysql> CREATE TABLE Department
    -> (
    -> DeptID INT PRIMARY KEY,
    -> DeptName VARCHAR(30)
    -> );
Query OK, 0 rows affected (0.09 sec)

mysql>
mysql> INSERT INTO Department VALUES
    -> (101,'HR'),
    -> (102,'IT'),
    -> (103,'Finance');
Query OK, 3 rows affected (0.02 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> CREATE VIEW DeptSalaryView AS
    -> SELECT D.DeptName,
    -> COUNT(E.EmpID) AS TotalEmployees,
    -> AVG(E.Salary) AS AverageSalary
    -> FROM Department D
    -> JOIN Employee E
    -> ON D.DeptID=E.DeptID
    -> GROUP BY D.DeptName;
Query OK, 0 rows affected (0.09 sec)

mysql> SELECT * FROM DeptSalaryView;
+----------+----------------+---------------+
| DeptName | TotalEmployees | AverageSalary |
+----------+----------------+---------------+
| HR       |              3 |    55000.0000 |
| IT       |              2 |    57500.0000 |
| Finance  |              2 |    35000.0000 |
+----------+----------------+---------------+
3 rows in set (0.07 sec)

mysql>
```

7Q)

create database studentdb;
use studentdb;

create table student(
sid int primary key,
name varchar(30)
);

create table course(
courseid int primary key,
coursename varchar(30)
);

create table enrollment(
sid int,
courseid int,
primary key(sid,courseid),
foreign key(sid) references student(sid),
foreign key(courseid) references course(courseid)
);

insert into student values
(1,'rahul'),
(2,'sneha'),
(3,'arjun');

insert into course values
(101,'dbms'),
(102,'java'),
(103,'python');

insert into enrollment values
(1,101),
(1,102),
(1,103),
(2,101),
(2,102),
(3,101);

OUTPUT :

```
3 rows in set (0.07 sec)

mysql> INSERT INTO Employee
    -> VALUES(8,'Kavya',65000,102);
Query OK, 1 row affected (0.05 sec)

mysql> UPDATE Employee
    -> SET Salary=70000
    -> WHERE EmpID=8;
Query OK, 1 row affected (0.05 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> DELETE FROM Employee
    -> WHERE EmpID=8;
Query OK, 1 row affected (0.05 sec)

mysql> |
```

8Q)
use companydb;

insert into employee
values(6,'riya',50000,102);

select * from employee;

update employee
set salary=65000
where empid=6;

select * from employee;

delete from employee
where empid=6;

select * from employee;

OUTPUT :

```
mysql> INSERT INTO Employee
    -> VALUES(8,'Kavya',65000,102);
Query OK, 1 row affected (0.06 sec)

mysql> UPDATE Employee
    -> SET Salary=70000
    -> WHERE EmpID=8;
Query OK, 1 row affected (0.05 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> DELETE FROM Employee
    -> WHERE EmpID=8;
Query OK, 1 row affected (0.05 sec)

mysql> |
```

9Q)

```
use companydb;

create table department1(
deptid int primary key,
deptname varchar(30)
);

insert into department1 values
(101,'hr'),
(102,'it'),
(103,'finance');

select * from employee;
select * from department1;

select e.empid,
e.empname,
e.salary,
d.deptname
from employee e
join department1 d
on e.deptid=d.deptid
where d.deptname='it'
and e.salary>50000;
```

OUTPUT :

```
    -> Employee(E)
    -> ∧ E.DeptID = 102
    -> ∧ E.Salary > 50000
    -> };
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near '{ E |
Employee(E)
? E.DeptID = 102
? E.Salary > 50000
}' at line 1
mysql> SELECT E.EmpID,
    ->        E.EmpName,
    ->        E.Salary,
    ->        D.DeptName
    -> FROM Employee E
    -> JOIN Department D
    -> ON E.DeptID = D.DeptID
    -> WHERE D.DeptName = 'IT'
    -> AND E.Salary > 50000;
+-------+---------+--------+----------+
| EmpID | EmpName | Salary | DeptName |
+-------+---------+--------+----------+
|     2 | Sneha   |  60000 | IT       |
|     3 | Arjun   |  55000 | IT       |
+-------+---------+--------+----------+
2 rows in set (0.03 sec)

mysql> |
```

10Q)

```
create database airlinedb;
use airlinedb;
```

```sql
create table passenger(
passengerid int primary key,
passengername varchar(30),
phone varchar(15)
);

create table flight(
flightid int primary key,
flightname varchar(30),
source varchar(30),
destination varchar(30)
);

create table booking(
bookingid int primary key,
passengerid int,
flightid int,
journeydate date,
foreign key(passengerid) references passenger(passengerid),
foreign key(flightid) references flight(flightid)
);

insert into passenger values
(101,'rahul','9876543210'),
(102,'sneha','9876543211'),
(103,'arjun','9876543212'),
(104,'priya','9876543213');

insert into flight values
(201,'air india','chennai','delhi'),
(202,'indigo','bangalore','mumbai'),
(203,'spicejet','hyderabad','kolkata');

insert into booking values
(301,101,201,'2026-08-10'),
(302,102,202,'2026-08-11'),
(303,103,203,'2026-08-12'),
(304,104,201,'2026-08-13');

select * from passenger;
select * from flight;
select * from booking;
```

```
select passengername,flightname
from passenger p
join booking b
on p.passengerid=b.passengerid
join flight f
on b.flightid=f.flightid;

select passengername
from passenger
where passengerid in
(
select passengerid
from booking
);

select flightname,passengername
from flight f
join booking b
on f.flightid=b.flightid
join passenger p
on p.passengerid=b.passengerid;

create view flightbookingview as
select passengername,flightname
from passenger p
join booking b
on p.passengerid=b.passengerid
join flight f
on b.flightid=f.flightid;

select * from flightbookingview;

select flightid,
count(*) as totalbookings
from booking
group by flightid;
```

OUTPUT :

```
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> INSERT INTO Booking VALUES
    -> (301,101,201,'2026-08-10'),
    -> (302,102,202,'2026-08-11'),
    -> (303,103,203,'2026-08-12'),
    -> (304,104,201,'2026-08-13');
Query OK, 4 rows affected (0.01 sec)
Records: 4  Duplicates: 0  Warnings: 0

mysql> show table
    -> show table;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'show table' at line 2
mysql> show table ;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near '' at line 1
mysql> SELECT * FROM Passenger;
+-------------+---------------+------------+
| PassengerID | PassengerName | Phone      |
+-------------+---------------+------------+
|         101 | Rahul         | 9876543210 |
|         102 | Sneha         | 9876543211 |
|         103 | Arjun         | 9876543212 |
|         104 | Priya         | 9876543213 |
+-------------+---------------+------------+
4 rows in set (0.00 sec)

mysql>
mysql> SELECT * FROM Flight;
+----------+------------+-----------+-------------+
| FlightID | FlightName | Source    | Destination |
+----------+------------+-----------+-------------+
|      201 | Air India  | Chennai   | Delhi       |
|      202 | Indigo     | Bangalore | Mumbai      |
|      203 | SpiceJet   | Hyderabad | Kolkata     |
+----------+------------+-----------+-------------+
3 rows in set (0.00 sec)

mysql>
mysql> SELECT * FROM Booking;
+-----------+-------------+----------+-------------+
| BookingID | PassengerID | FlightID | JourneyDate |
+-----------+-------------+----------+-------------+
|       301 |         101 |      201 | 2026-08-10  |
|       302 |         102 |      202 | 2026-08-11  |
|       303 |         103 |      203 | 2026-08-12  |
|       304 |         104 |      201 | 2026-08-13  |
+-----------+-------------+----------+-------------+
4 rows in set (0.00 sec)

mysql> |
```